

are all good examples. Automobiles, especially two-wheeler EVs, is another fast growing segment. Another example is security cameras with built-in AI capabilities. These are all large-volume markets in India that can provide semiconductor companies an opportunity to develop India-centric chips and carve out a niche.

How do you see edge computing playing a role in set-top boxes?

IESA (Hyderabad chapter) recently hosted a two-day summit on artificial intelligence. Edge computing was an important topic during the summit. The consensus is that every electronic device on the edge is going to have some AI built into it. A set-top at home can create a personalised user experience with built-in AI capabilities.

What is your view on the RISC-V phenomenon that is becoming a rage in the chip-design eco-system? What are your views on it?

I think it is gaining a lot of momentum. At MosChip, we are also quite excited about opportunities RISC-V can open for us, especially in India. ARM has been the leading processor for mobile and IoT applications and I think they have been making good progress into server market as well. ARM is in the process of getting acquired by Nvidia. Countries like India and China are focused on self-reliance and have started investing in RISC-V.

A few weeks ago I had an interesting meeting with Professor Kamakoti of IIT Madras. (The timing is a nice coincidence wrt what we are talking about.) IIT Madras has been doing good work and investing a lot in developing Shakti processor. In parallel, MosChip has been developing an edge-computing platform we do see opportunities to collaborate with IIT, Madras.

A lot of work still needs to be done to make RISC-V a viable alternative to ARM. MosChip will work closely with IIT Madras and support Indian government initiatives towards indigenous processor platform.

For local semicon majors to grow, many claim that we need a fab in India. What's your take on this issue—should India invest?

If you look at the whole supply chain, most of the chip companies are now fabless companies, including companies like AMD, Broadcom, and Qualcomm. None of them have their own fabs. Most of them get their chips fabricated at TSMC, GlobalFoundries,

and others. And the packaging and testing of the chips is done through other partners. Packaging and testing of the chip (usually called ATMP or OSAT) addresses 30% to 40% of the chip cost.

ATMP facility, which does not cost as much as a fab, can help create a local eco-system and support local fabless start-ups or multinationals operating out of India.

With a lot of buzz around shortage of semiconductors, what are the opportunities opening up for India, if any?

The Indian government has invited 'Expression of Interest' to set up fabs in India. I do see opportunities in some specialty fabs, but it will take time for India to make significant contribution. And it needs a focused effort for the next ten years and sustained efforts thereafter.

How do you see the evolution of Moore's Law? Will it remain relevant?

It has remained relevant for longer than the industry anticipated. In my own career, I have seen the transistor shrinking from 1 micron to 3nm. The hunger for more computing will continue to innovate new transistor structures and new technologies like quantum computing.

While semiconductors are powering AI and ML, are AI and ML enabling R&D, development, and manufacturing of chips too? Are they going to play an important role in the future?

AI and ML certainly created R&D opportunities to develop new architectures for handling the special computing and data processing needs of AI applications. AI is changing the way the data is stored, processed, and handled (security). I do see AI playing a role in semiconductor manufacturing as well, to improve both efficiency and reliability.

Can you share some details about the MosChip Institute of Silicon System? Who does it target—freshers or industry professionals? And does having your own institute help in getting industry-ready talent without having to invest in training them?

Training has been my passion since 1999, when I was running TTM (Time to Market Inc). We were the first company in India to offer Physical Design training. The team from the Institute of Silicon Systems has recreated and done a better job; the institute is now a part of MosChip. We are endorsed as the best Physical Design training institute in India currently, and we offer similar

training in RTL Design/Verification, Analog Design, and Embedded System Design.

Most of the engineers who come to our institute are freshers and the training prepares them to start contributing in design, with some handholding from seniors. The institute certainly helps us in recruiting the talent and, in fact, we let other product and design services companies also recruit from the training institute.

There was a recent announcement with respect to a focus on turnkey ASICs. What has changed or is going to change now?

When MosChip was started around 1999, it was actually a product company. It was the first fabless semiconductor company out of India, designing and selling their own chips. The chips were mostly connectivity based for USB, PCI, Ethernet, etc. MosChip got the chips fabricated mostly at UMC and sold them in millions. So, it was a true product company selling chips with their own logo. But, unfortunately, for various reasons, the company had to divest that group and started focusing on the design services.

When I took over the role of CEO at MosChip, I evaluated the strengths of the organisation and came with a long-term strategy. Turnkey ASIC was one area we really wanted to focus on. Turnkey ASIC allows us to build and sell products instead of pure play design services. MosChip has done turnkey ASICs for the Indian government with an ARM processor and other connectivity interfaces. The ASIC was actually fabricated at TSMC and we delivered the working parts to the Indian government; the ASICs were targeted towards handheld applications.

The expertise of designing an ASIC and taking into production remained with MosChip, helping us with the turnkey ASIC strategy.

Am I correct to understand that rather than just be a design house, you now want MosChip to be an ODM for ASICs?

Yes, right now we are already giving the design to the customer. So going forward, we will actually give the chip to the customer. We are already doing that with a small number of customers, but we want to focus on large-volume customers. **EFY**

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